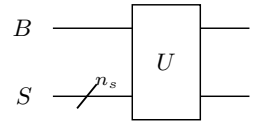
(a) Block encoding U of X (dilation; `QSVT._walk_power`)

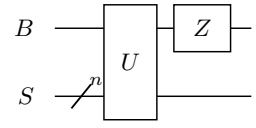


$$U = \begin{pmatrix} X & \sqrt{I-X^2} \\ \sqrt{I-X^2} & -X \end{pmatrix}, \quad (\langle 0|_B \otimes I)\, U\, (|0\rangle_B \otimes I) = X$$

$$X = \frac{2A - (d_{lo}+d_{hi})I}{d_{hi}-d_{lo}}, \text{ spectrum in } [-1,1];\ x_j = \cos\theta_j$$

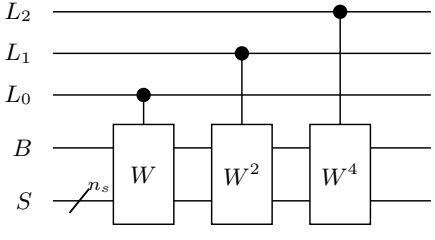
Exact and real symmetric, but dense: the package builds it as a matrix (a `UnitaryGate`). A deployable circuit would come from sparse-access oracles; 7.3 and paper 10.8 price that as a model ($103\,T^{0.85}$ two-qubit gates per call), not a built circuit.

(b) Walk step $W = (Z_B \otimes I)\, U$



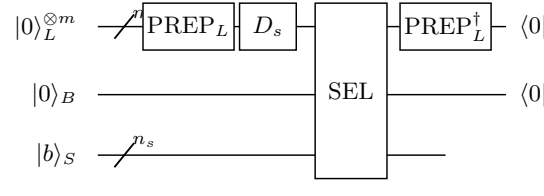
In each plane $\{|0\rangle_B|v_j\rangle, |1\rangle_B|v_j\rangle\}$, W is a rotation by θ_j:
$W^k|0\rangle_B|b\rangle = |0\rangle_B\, T_k(X)|b\rangle - |1\rangle_B\, S_k|b\rangle$

(c) SELECT $= \sum_k |k\rangle\langle k|_L \otimes W^k$ by binary powering ($m = 3$ drawn)



$c\text{-}W^{2^j}$ from index qubit L_j (`_walk_power(1<<j)`). $|k\rangle_L$ then carries W^k for $k = \sum_j k_j 2^j$: $2^m - 1$ walk steps in all (d with an optimal SELECT).

(d) LCU: PREP, SELECT, PREP†, post-select



$$(\langle 0|_L\langle 0|_B)\,|\Phi\rangle = \frac{1}{\|c\|_1}\sum_k c_k T_k(X)|b\rangle = \frac{p(X)|b\rangle}{\|c\|_1}$$

$$P = \|p(X)b\|^2 / \|c\|_1^2$$

Registers as in `QSVT.build_circuit` (little-endian): system S (n_s) | block-encoding ancilla B (1) | index L ($m = \lceil \log_2(d+1) \rceil$). PREP_L and D_s are C1c. The package code applies $\text{PREP}_L^\dagger$ without $D_s^\dagger$: on the post-selected $|0\rangle_L$ branch the sign is already folded into c_k. `solve_statevector` computes the same post-selected state by the recursion $T_{k+1} = 2XT_k - T_{k-1}$ (agrees with `run_circuit` to 2×10^{-15}).